Sikao Guo, Ph.D.

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Computational Scientist | Scientific Software Engineer | No Sponsorship Required

PROFESSIONAL SUMMARY

Computational biophysicist and scientific software engineer with 6+ years of experience developing mechanistic biological models and reusable computational workflows. Builds C++ and Python software spanning stochastic reaction-diffusion, state-transition modeling, model generation, quantitative validation, MPI/HPC, and scientist-facing workflow infrastructure. Experienced in translating biological mechanisms and experimental measurements into computational model states, parameters, observables, and testable predictions.

EXPERIENCE

Inria | Research Engineer

Jul. 2025 — Present | Sophia Antipolis, France

- Architected a specification-driven Python workflow that converts scientific application metadata into validated, platform-independent JSON representations and automatically generates consistent VMD, PyMOL, and web applications, separating model state, execution logic, interfaces, and post-analysis.
- Developed C++ ensemble-sampling methods for protein backbones and intrinsically disordered regions using inverse kinematics, released endpoint constraints, SE(3) Hit-and-Run proposals, multi-frame resampling, and spatial-hash collision detection.
- Built a reproducible ensemble validation and selection workflow integrating side-chain completion, OpenMM scoring/minimization, probabilistic conformational validation, MDS/clustering, quality gates, and deterministic representative selection; benchmarked generated ensembles against BBFlow and idpGAN.

Johns Hopkins University | Postdoctoral Researcher, Assistant Research Scientist

Jan. 2020 — Jun. 2025 | Baltimore, MD

- Designed and implemented MPI spatial domain decomposition and ghost-region communication for the C++ NERDSS simulator, scaling particle-based reaction-diffusion and self-assembly simulations to 96 CPUs.
- Developed structure-resolved stochastic models of clathrin-mediated endocytosis and HIV-1 Gag assembly by integrating structural, kinetic, and experimental data into calibrated computational models.
- Predicted an adaptor-to-clathrin threshold above 1:1, a critical nucleus of approximately 25 clathrin trimers, and minute-scale Gag-Pol dimerization, connecting model states and kinetic parameters to experimentally testable biological behavior.
- Co-developed ioNERDSS, an open-source Python workflow that converts PDB/mmCIF structures into simulation-ready coarse-grained reaction-diffusion models, integrating interface detection, repeated-subunit regularization, reaction enumeration, baseline kinetics, and ML-assisted affinity initialization.
- Designed the workflow to translate structural information into explicit model components, reactions, parameters, and simulation artifacts for reproducible downstream analysis.

Institute of Physics, Chinese Academy of Sciences | Doctoral Researcher

Sept. 2014 — Dec. 2019 | Beijing, China

- Developed kinetic/state-transition models of kinesin motors that quantitatively reproduced force- and ATP-dependent velocity, dwell time, run length, and processivity across diverse single-molecule experiments.
- Connected molecular conformational changes and chemical reaction rates to emergent motor behavior through mathematical modeling, simulation, and quantitative comparison with experimental data.
- Conducted molecular-dynamics simulations of kinesin–microtubule interactions to test mechanistic hypotheses used in the kinetic models.

EDUCATION

Institute of Physics, Chinese Academy of Sciences | Ph.D. Physics

Sept. 2014 — Dec. 2019 | Beijing, China

Nankai University | B.S. Physics

Sept. 2010 — Jun. 2014 | Tianjin, China

OPEN-SOURCE SOFTWARE

Software and project details: [Project Portfolio]

Scientific GUI Framework (GitHub | Documentation) *Cross-platform scientific GUI generation framework.*

NERDSS-MPI (GitHub | Documentation) *Parallel C++ protein assembly simulator using MPI.*

ioNERDSS (GitHub | Documentation) *Automated construction of coarse-grained reaction--diffusion models.*

SELECTED PUBLICATIONS

Full publication list: Google Scholar

Guo, S., Sarti, E., and Cazals, F. "A Unified, Cross-Platform Framework for Automatic GUI and Plugin Generation in Structural Bioinformatics and Beyond". *arXiv*, 2026. arXiv:2602.16047.

Guo, S., Korolija, N., et al. "Parallelization of Particle-Based Reaction--Diffusion Simulations Using MPI". *Journal of Computational Chemistry*, 2025. doi:10.1002/jcc.70132.

Ying, Y. M., Sang, M., ..., **Guo, S.**, et al. "Transforming Macromolecular Structures into Simulations of Self-Assembly with ioNERDSS". *bioRxiv*, 2026. doi:10.64898/2026.01.27.702082.

Guo, S. et al. "Structure of the HIV Immature Lattice Allows for Essential Lattice Remodeling within Budded Virions". *eLife*, 2023. doi:10.7554/eLife.84881.

Guo, S., Sodt, A. J., Johnson, M. E. "Large Self-Assembled Clathrin Lattices Spontaneously Disassemble without Sufficient Adaptor Proteins". *PLOS Computational Biology*, 2022. doi:10.1371/journal.pcbi.1009969.

TECHNICAL STRENGTHS

Programming & scientific computing: Python, C++, MPI, Bash, Linux, Git, NumPy/SciPy, parallel algorithms

Mechanistic modeling: stochastic reaction-diffusion, state-transition models, kinetic modeling, ensemble simulation, coarse-graining, molecular simulation

Workflow & model infrastructure: reproducible pipelines, model generation, quantitative validation, automated testing concepts, JSON Schema, scientific code generation, HPC

Structural & computational biology: protein modeling, biomolecular assembly, PDB/mmCIF, OpenMM, VMD, PyMOL